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# Centerline

TECHNICAL SERVICES CO. LLC

Turnkey

Healthcare Infrastructure Solution Provider

## COMPANY PROFILE

THE NO.1 CHOICE

www.centerline.ae

## MESSAGE FROM MANAGEMENT

At Centerline, we are committed to transforming healthcare infrastructure to meet the evolving needs of modern medical care. With a clear mission to enhance efficiency, safety, and adaptability in healthcare facilities, we specialize in delivering innovative solutions in healthcare ventilation, modular construction, electrical practices, and infection control systems.

We understand the unique challenges healthcare providers face – from ensuring optimal air quality and reliable power systems to adapting to rapidly changing medical needs while maintaining compliance with stringent regulations. Our team is dedicated to addressing these challenges through cutting-edge technologies and tailored solutions that prioritize patient care, safety, and operational excellence.

With a focus on innovation and excellence, Centerline continues to set new standards in healthcare infrastructure services. Our infection control systems, advanced HVAC solutions, modular construction methods, and robust electrical practices are designed to mitigate risks, protect patients and staff, and support the seamless integration of modern medical technologies.

We are proud to support the healthcare industry with solutions that drive better outcomes and safer environments. As we move forward, we remain committed to driving innovation, excellence, and reliability in all that we do, setting new benchmarks in healthcare infrastructure services.

Sincerely,
Management Team, Centerline

## OUR HISTORY

Founded in 2024, Centerline was established to address the critical infrastructural needs of the healthcare environment. With a mission to enhance healthcare facilities' efficiency, safety, and adaptability, the company specializes in healthcare ventilation, modular construction, electrical practices, and infection control systems. From its inception, Centerline aimed to address challenges faced by healthcare providers, including outdated infrastructure, compliance with stringent regulations, and the need for advanced, patient-centric environments. With a vision to support the healthcare industry through innovative solutions, the company set out to build a strong foundation for safer, more efficient, and adaptive healthcare spaces.
Recognizing the growing importance of infection prevention in healthcare, Centerline integrates advanced technologies and methodologies to develop state-of-the-art infection control systems. These solutions help mitigate risks, protect patients and staff, and ensure compliance with stringent health and safety standards.

Recognizing the unique challenges faced by healthcare providers—such as maintaining optimal air quality, ensuring reliable power systems, and adapting to rapidly evolving medical needs, Centerline positioned itself as a leader in delivering tailored solutions. From designing advanced ventilation systems that promote clean, safe air to pioneering modular construction methods for rapid and cost-effective facility development, the company has made significant strides in supporting the healthcare sector. Additionally, its expertise in electrical practices ensures the seamless integration of modern medical technologies and robust power infrastructure. From designing advanced ventilation systems that promote clean, safe air to pioneering modular construction methods for rapid and cost-effective facility development, Centerline has made significant strides in supporting the healthcare sector. Its expertise in electrical practices ensures the seamless integration of modern medical technologies and robust power infrastructure, while its infection control systems set new standards for safety and reliability.

Centerline continues to drive innovation and excellence, setting new benchmarks in healthcare infrastructure services.

## ABOUT US

At Centerline Technical Services, we specialize in delivering complete turnkey solutions tailored for the healthcare industry and beyond. As the medical equipment and infrastructure market continues to evolve, we recognize the importance of integrated systems that ensure safety, efficiency, and high-quality patient care.

Our project management team works with precision and dedication to meet client expectations, providing innovative, value-added products and services that improve community well-being. With a strong foundation in knowledge, experience, and global best practices, we deliver reliable solutions that empower hospitals, laboratories, doctors, and patients alike. We design, develop, supply, install, test, commission, and maintain state-of-the-art healthcare infrastructure solutions, aligning with the latest technologies and industry standards.

## WHY CHOOSE US

- End-to-end turnkey healthcare infrastructure solutions.
- Deep industry expertise with latest technology integration.
- Commitment to safety, reliability, and compliance.
- Proven project management excellence.
- Enhancing patient care and healthcare facility efficiency.

# OUR SERVICES

## HEALTHCARE HVAC SYSTEMS

## HEALTHCARE MODULAR CONSTRUCTION

## INFECTION CONTROL SYSTEMS

## MEDICAL GAS SUPPLY SYSTEMS (BHU & PENDANTS)

## HOSPITAL ELECTRICAL SYSTEMS

# HEALTHCARE HVAC SYSTEMS

The need for specialized Healthcare HVAC (Heating, Ventilation, and Air Conditioning) is critical and non-negotiable in modern medical facilities. It goes far beyond simple comfort cooling and heating; it is a first-line defence mechanism for infection control, patient safety, and the success of medical procedures.

The need for healthcare HVAC is a fundamental requirement for patient safety, clinical efficacy, and operational integrity. It is a sophisticated engineering discipline that integrates mechanical systems with medical science and infection control protocols. Investing in a properly designed, installed, and maintained healthcare HVAC system is not just about climate control; it is an direct investment in positive patient outcomes and saving lives.

## CORE OBJECTIVES OF HEALTHCARE HVAC

- Protect Staff and Visitors
- Protect Patients
- Enable Precision Medicine
- Operational Continuity
- Protect the Surrounding

# APPLICATION BY MODALITIES

### Operating Rooms

Positive Pressure
Laminar flow units
HEPA filters
22 - 25 ACH
Velocity : 0.25 - 0.45m/s

### Isolation Rooms
(Infectious)

Negitive Pressure
Extact air HEPA & UV
12 ACH
Differential Pressure Monitoring

### P.E Rooms
(Immunocompromised)

Positive Pressure
HEPA filter
Differential Pressure Monitoring

### Pharmacies

Hazardous (Negative Pressure)
Non-Haz (Positive Pressure)
HEPA filter
USP 797 ISO Class 7

### Laboratories

Positive & Negative Pressure
HEPA filters
Differential Pressure Monitoring

### Imaging Hybrid ORs

Positive Pressure
HEPA filters
Differential Pressure Monitoring

## HEALTHCARE MODULAR CONSTRUCTION

The need for healthcare modular construction is more pressing than ever. It addresses critical challenges in the healthcare industry by offering a faster, more efficient, and often superior alternative to traditional building methods.

### EXECUTIVE SUMMARY: THE PRESSING NEED

The healthcare sector faces immense pressure from aging populations, rising costs, rapidly evolving technology, and the constant threat of public health crises (like pandemics). Traditional construction is often too slow, too disruptive, and too expensive to meet these demands. Modular construction emerges as a vital solution, offering speed, precision, cost certainty, and scalability that modern healthcare requires.

| | |
|---|---|
| Patient Tower Additions | Multi-Storey wings with patient rooms and nurse stations. |
| Emergency Departments (ED) and Urgent Care Centers: | Fast-deployment units to expand capacity. |
| Diagnostic and Imaging Suites: | MRI, CT scan, and X-ray rooms requiring precise shielding. |
| Laboratory Spaces | Including complex BSL-2 (Biosafety Level 2) labs for research and testing. |
| Pharmacy Facilities | Sterile compounding rooms and retail pharmacies. |
| Surgical Suites | Operating rooms built to exacting standards for air filtration, lighting, and equipment integration. |
| Clinical Offices & Administration Buildings: | Separate structures for outpatient clinics and administrative staff. |
| Mobilization and Temporary Facilities | Surge capacity units, temporary testing or vaccination clinics, and swing space during renovations. |

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# INFECTION CONTROL SYSTEMS

Infection control infrastructure is not a single entity, but a complex, multi-layered system integrating physical design, engineering, equipment, protocols, and human resources. Its goal is to break the chain of infection and protect patients, staff, and visitors.

## PATIENT ROOM DESIGN:

1. Contaminating themselves or surfaces.
2. Dedicated Hand Hygiene Sinks
3. Architectural design as per JCI requirement to curb infection.

## VENTILATION AND AIR HANDLING SYSTEMS (HVAC):

## NEGATIVE PRESSURE ROOMS

Airborne Infection Isolation Rooms (AIIRs or Negative Pressure Rooms):
For patients with diseases like tuberculosis, measles, or COVID-19.
These rooms have:

1. Negative pressure (air flows into the room from the corridor).
2. ≥12 air changes per hour (ACH).
3. HEPA-filtered exhaust to the outside or recirculated after HEPA filtration.

## PROTECTIVE ENVIRONMENT (PE) ROOMS:

1. For highly immunocompromised patients (e.g., bone marrow transplant).
   These have:
2. Positive pressure (air flows out of the room to the corridor).
3. HEPA-filtered supply air.
4. Sealed rooms to maintain pressure differentials.

## UNI-DIRECTIONAL AIRFLOW IN KEY DEPARTMENTS:

Operating rooms, ICUs, and procedural suites require controlled, clean-to-dirty airflow patterns.

Non-porous, Seamless Surfaces: Floors, walls, and countertops made of materials that can withstand frequent and aggressive cleaning with disinfectants (e.g., epoxy resin floors, solid surface counters).

Copper-Alloy Surfaces: Increasingly used on high-touch items (bedrails, IV poles, door handles) for their intrinsic antimicrobial properties.
Sinks with Hands-Free Operation: Sensor-activated or foot-pedal controlled to avoid recontamination.

## WATER SAFETY SYSTEMS:

### Legionella Prevention:

Water management plans that include routine flushing, temperature control, and sometimes copper-silver ionization or chlorine dioxide systems to prevent biofilm and bacterial growth.

## ULTRAVIOLET (UV) DISINFECTION SYSTEMS:

UV-C Robots: Used after terminal cleaning in operating rooms and patient rooms to decontaminate surfaces and air by damaging microbial DNA.

UV-C in HVAC Units: To disinfect air as it circulates.

## NEEDLESTICK AND SHARPS INJURY PREVENTION:

Safety-engineered devices (e.g., self-sheathing needles, retractable scalpels) and strategically placed sharps containers.

## ELECTRONIC HAND HYGIENE MONITORING:

Systems that use sensors (on dispensers, badges) to provide feedback on compliance rates, replacing unreliable human auditors.

## REAL-TIME LOCATION SYSTEMS (RTLS):

To track equipment cleaning status, monitor patient movements for exposure tracing, and ensure isolation compliance.

## TELEHEALTH CAPABILITIES:

For remote consultations with patients in isolation, reducing unnecessary exposures and conserving PPE.


# MEDICAL GAS SUPPLY UNITS (BHU & PENDANTS)

Hospital bed head units and pendants are critical components of patient room design, delivering essential medical gases, power, and data to the point of care.

Both Bed Head Units (BHUs) and Pendants are integrated wall-mounted or ceiling-mounted systems designed to consolidate and deliver a suite of services to a hospital bed or operating table. Their primary purpose is to remove clutter and trip hazards from the floor, improve patient and staff safety, and provide reliable access to critical utilities.

## CORE SERVICES PROVIDED BY BHU AND PENDANTS

### MEDICAL GASES:

Oxygen ($O_2$), Suction (Vac), Medical Air (Med Air), Nitrogen ($N_2$), Nitrous Oxide ($N_2O$), AGSS and $CO_2$

### ELECTRICAL POWER:

Numerous AC outlets for medical equipment (ventilators, infusion pumps, monitors) and personal devices.

### DATA AND COMMUNICATION:

Numerous AC outlets for medical equipment (ventilators, infusion pumps, monitors) and personal devices.

### LIGHTING:

Often integrated examination lights or reading lights.

## PROS:

### SUPERIOR FLEXIBILITY & ACCESS:

The articulating arms can be positioned exactly where needed, providing perfect ergonomics for staff. The bed can be placed anywhere in the room.

### CLEAR FLOOR SPACE:

Eliminates all cables and hoses from the floor, greatly improving safety and cleanliness.

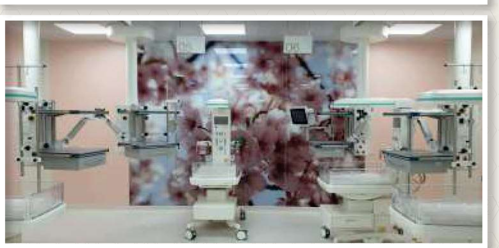
### HIGH CAPACITY:

Can be configured with a very large number of outlets and services.

### STORAGE:

Medical equipment hanging on the rails the better access the the caregiver.

| | |
|---|---|
| Modular Design | Allows you to configure the exact combination of gas outlets, electrical sockets, and data ports you need. Also makes future upgrades or changes much easier. |
| Touchless/ Smart Controls: | Modern units may feature touchless activation of lights or nurse call via motion sensors or voice control to enhance infection control. |
| LED Lighting: | Integrated, energy-efficient LED reading lights and examination lights with adjustable color temperature (e.g., warm for patient comfort, cool for clinical assessment). |
| Connectivity: | USB charging ports for patients and staff are now a standard expectation. |
| IP Rating (Ingress Protection): | A rating like IP54 signifies protection against dust and water jets, making the unit suitable for harsher environments like ICUs or ORs where frequent cleaning with liquids occurs. |
| Aesthetics: | Modern designs focus on a clean, minimalist look to create a less clinical and more calming patient environment. Color options are often available. |



# HOSPITAL ELECTRICAL SYSTEM

Unlike a standard commercial building, a hospital's electrical system is a critical life-support system. A power outage is not just an inconvenience; it can directly impact patient survival. Therefore, the entire system is designed around redundancy and reliability.

A hospital's electrical system is a marvel of engineering designed for one purpose: to never fail. It is a multi-layered, redundant, and rigorously tested system that ensures continuous power to life-saving equipment, protecting both patients and staff under the most extreme circumstances. The design philosophy is not if a component fails, but when it fails, the next layer of backup is already in place to take over.

## KEY COMPONENTS OF THE SYSTEM

A hospital's electrical infrastructure is a layered system of backups:

1. Normal Power Source: The primary electricity feed from the local utility grid.
2. Automatic Transfer Switches (ATS): The brain of the backup system. They constantly monitor the normal power. If it fails, they automatically and seamlessly (within 10 seconds as required by code) switch the load to the emergency source.
3. Emergency Power Source:

   ∗Generators:
   ∗UPS (Uninterruptible Power Supply):

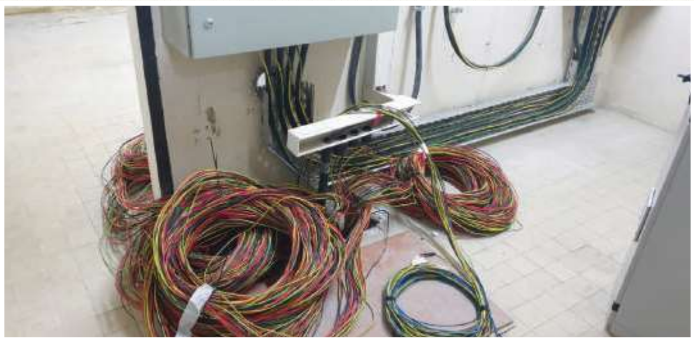
# KEY COMPONENTS OF THE SYSTEM

## SPECIALIZED SYSTEMS

Grounding: Perhaps the most critical aspect. Hospitals use a "Technical Ground" or "Patient Care Grounding" system. This is a separate, ultra-low impedance grounding system to ensure all exposed metal surfaces and equipment are at the same electrical potential. This prevents microshock hazard—a tiny, imperceptible current that could be fatal to a patient with a direct cardiac connection (like a catheter).

## ISOLATED POWER SYSTEMS (IPS):

Used in wet locations (OR, Cath Labs, ICU) where the risk of leakage current is high. An IPS uses an isolation transformer to decouple the power circuit from the ground. A Line Isolation Monitor (LIM) continuously checks the integrity of the system and alarms if excessive leakage is detected, indicating a potential hazard, but without shutting off power—which is critical during surgery.



# OUR CLIENTELE

Ministry of Health UAE
(Dhaid Hospital)

EHS مؤسسة الإمارات للخدمات الصحية
Emirates Health Services

برجيل
burjeel
by Burjeel Holdings

ART

MUBADARAT
The President's Initiatives
Sheikh Khalifa Hospital
Fujairah
مبادرات
رئيس الدولة
مستشفى الشيخ خليفة
الفجيرة

Elyzee
Hospital

ميديكلينيك
MEDICLINIC

وزارة الصحة
Ministry of Health

ROYAL HEALTH
RHG

Ministry of Health UAE
(Al Qassimi Hospital)

EHS مؤسسة الإمارات للخدمات الصحية
Emirates Health Services

... AND MORE...

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